

# Chapter 14

## An Overview of Major Fungal Diseases of Sugarcane in India: Detection and Management Strategies

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### 14.1 Introduction

Sugarcane is one of the most important commercially crop grown in more than 115 tropical and sub tropical countries of the world. India is a major producer as well as consumer of sugar in the world and produced about 25 metric tons (MT) of sugar from 360 MT of sugarcane in year 2011–2013, contributing about 15 % of the totals sugar production of the world. A quantum of sugar is produced from sugarcane; however, this crop faces a number of problems such as low cane productivity, biotic and abiotic stresses, high cost of cultivation, unavailability of seed cane of newly released varieties, post harvest losses and low sugar recovery (Tiwari et al. 2015a). Sugarcane is long duration vegetative propagated crop and is being attacked by a number of pathogens (Fungi, Bacteria, Phytoplasma and Viruses) according to the prevailing climatic conditions from germination till maturity. About 60 diseases

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have been reported in India to attack this crop. Out of these about dozen diseases are of major and economically important. Most of the major diseases are seed piece transmissible and inter state transportation of seed material without any quarantine system results in the spread of these diseases. On an average the loss caused in sugarcane by fungal diseases alone ranges from 15 to 30 % (Tiwari et al. 2010). But in case of epiphytotic conditions the losses in yield and sugar goes up to 100 % (Viswanathan and Rao 2011). The fungal diseases are comparatively more damaging than others. In this chapter, fungal disease and their impact on sugarcane has been described.

## 14.2 Major Fungal Diseases of Sugarcane

### 14.2.1 Red Rot of Sugarcane

The term sugarcane and red rot diseases of sugarcane go hand in hand. Even in early eighth to seventh century BC the disease was mentioned in the Vedic text of Aitereya Aryanaka. Where it is stated that in Indian epic Mahabharata there was a Pundra kingdom where the people suffered from a skin disease called Pandu and it is said the name Pandu originated from a sugarcane variety then found in that region which was affected by a diseases of red patches. One also finds the mention of red rot in the Buddhist literature. The disease was so prevalent in UP and Bihar where Gautama Buddha started preaching after enlightenment. Buddha referred to red rot of sugarcane as Manjitthika after a red dye. Gautama Buddha quoted “just as the disease known as Manjitthika falls on a field of ripened sugarcane that field does not last long”. In India the first red rot epidemics occurred in the Godavari delta in 1895. The disease devastated the entire sugarcane cultivation in the Godavari delta to the extent that area under sugarcane reduced to 500 hectare (ha) from more than 2500 ha by 1899. Hence, the government deputed Mr C.A Barber to study the crop and the disease. In his report he described the causal agent a fungal pathogen named as *Colletotrichum falcatum* Went (Panicker and Velazhahan 2015).

Red rot is most serious sett borne disease and also known as cancer of sugarcane. It has appeared many times in the form of epiphytotic in U.P., Bihar, Haryana, Punjab and Kerala. In Uttar Pradesh and Bihar (two major cultivars i.e. Co 213 and Co 312) were badly affected by this disease causing epiphytotic conditions in the year 1938–1940 and 1946–1947. A very recently, important and popular variety CoS 07250 was rejected and gone out of cultivation because of red rot disease found at various places of eastern Uttar Pradesh, India. It is one of the most widespread sugarcane diseases in the country and has been a constraint for sugarcane productivity for the past 100 years in India. The disease is responsible for the elimination of many commercial varieties in India in the earlier decades. Epidemics of the disease have been very common ever since its occurrence in India. Currently the disease occurs in all the sugarcane growing states in India except Karnataka and Maharashtra

states. Recently reports of red rot occurrence in CoC 671 in parts of Kolhapur and Solapur Districts in Maharashtra were reported. In the recent years we have witnessed break down of cvs CoS 8436, CoSe 95422, 92423 and BO138, the important commercial varieties of subtropical region to red rot. Severe damage to crop stand is found in these varieties due to disease epidemics in the states of Haryana, Uttar Pradesh and Bihar.

#### **14.2.1.1 Economic Importance of Red Rot Disease of Sugarcane**

The pathogen affects the economically valuable stalk tissues and even the limited infection can bring about drastic changes in the juice quality. The disease affected cane gives poor sugar recovery because of impaired sucrose metabolism. The red rot infection reduced total carbohydrates in the diseased canes and the reduction was more in the highly susceptible varieties (Agnihotri 1996). Moreover, the pathogen produces abundant quantities of acid invertases which break the sucrose into glucose and fructose which are consumed by the pathogen. Higher production of acid invertases in the highly susceptible varieties was recorded upon pathogen infection as compared to resistant varieties (De Silva et al. 1977). Pathogen infection also resulted in increased level of total soluble salts, acid, reducing sugars, gum and simultaneously decrease in pH, sucrose and purity of cane juice (Singh and Waraitch 1977). Increased activity of the enzyme invertase or inhibition of normal synthesis of carbohydrates is indicated by degradation of sucrose and increase in the levels of reducing sugars. Similar studies conducted at SBI, Coimbatore revealed that pathogen infection has drastically reduced brix, sucrose percentage, purity and CCS per cent in the diseased canes. The affected canes recorded 25–75 % reduction in sucrose content than the healthy canes (Viswanathan and Samiyappan 1999a, b). Studies of Satyavir et al. (2002) in Haryana revealed that red rot infection reduces 7.1–32.5 % in extraction, 7.4–38.7 % in polarity, 0.5–8.3 % in purity co-efficient, 7.8–39 % in commercial cane sugar and an increase of 19.2–40.95 % in reducing sugars. During the milling process, mixing of juice from healthy and diseased canes result in spoilage of entire juice due to inversion of sucrose. Similarly ‘jaggery’ setting will also be affected if red rot affected canes are crushed with healthy ones. Resistance to red rot in sugarcane varieties is not static; hence, practically all varieties under cultivation are susceptible. Once a resistant variety occupying large areas succumbs to the disease, extensive damage to cane cultivation occurs.

#### **14.2.1.2 Epidemiology and Symptom of Red Rot Disease**

Sett borne inoculum is considered to be the major mode of inoculum transmission. Planting the infected material is the prime source of inoculum for recurrent disease incidence. Post germination, infection on young emerging shoots is caused by the dormant mycelium present in the bud scales. In the field, the various sources of

pathogen transmission are midrib lesions, disease stalks, infected setts, crop debris and infested soil. The means of inoculum transmission depend on the season and the cane growing circumstances. These sources of inoculum initiate infections in young developing canes in the nearby fields. The secondary spread of the disease is affected by conidial dissemination through irrigation water, rain splashing, dew washings from mid rib lesions and dispersal by wind. Bore holes could also act as portals of entry. Singh and Lal (1996) trapped propagules or structures of *C. falcatum* other than ascospores from air and confirmed development of nodal infection in healthy plants in the vicinity of the diseased ones. The resting structures of the fungus such as appressoria, chlamydospores, thick-walled hyphae and setae play a vital role in the soil borne transmission. Crop debris comprised of diseased stalk or stubble pieces left after harvest provide a bulk of inoculum to the subsequent infection under optimal high humid conditions. Though the fungus is not a true soil borne pathogen, enough evidence suggests that debris borne inoculum have a role in perpetuating the fungal propagules (Viswanathan and Rao 2011).

Expression of the disease may vary depending upon nature of infection and prevailing environmental conditions (Satyavir 2003). In the early stages of infection, it is difficult to recognize the presence of the disease in the field as reddening of the internal tissues with interrupted red and white patches, the characteristic symptoms of the disease, develops on the stem only at later stages. Furthermore, latent infection occurs frequently, making visual diagnosis impossible (Nithya et al. 2012). The symptoms of this disease start from the month of July/August and found till the end of the crop. Third to fourth leaf of the top starts withering along the margins. Later on, the whole crown withers away. At initial stages purple colour develops on the nodes of cane, which turns to reddish-brown later. Midrib lesions are the major source of inoculum during the cane growing season. Typical symptoms are observed in the internodes of the stalk after splitting the cane longitudinally. Dull reddening of tissues with white spots are seen which omit alcoholic smell. Red rot affected canes are very easily broken from the nodes. Infected standing stalks serve as a greater inoculum source at later stages of the crop (Fig. 14.1).

The first symptoms of the disease are seen when the vegetative growth of the plant is stopped and sucrose formation begins, i.e., after rainy season. The pathogen, *C. falcatum* Went, can attack any part of the sugarcane plant; be it stalk, leaf, buds or roots. *C. falcatum* completes its life cycle on the sugarcane leaf and usually damage to leaf does not pose a serious threat to cane or cause much harm to the plant (Duttamajumder 2008). Discolorations of the leaves are the first symptom in the field. The spindle leaves (third and fourth leaf) display drying which withers away at the tips along the margins. This discoloration from tip to the base is continued till all the leaves of the crown wilt (Agnihotri 1996). Tiny reddish lesions occur on the upper surface of the lamina with minute red spots in both the directions of the upper surface of the midrib. Infection also resulted in change in the colour of the leaves that become straw colored in the center and dark reddish brown at the margins with the development of black acervuli. The infected leaves may break at the lesions and hang down (Sharma and Tamta 2015).

**Fig. 14.1** Sugarcane stalk (LS) showing the symptoms of red dot disease



#### 14.2.1.3 Molecular Markers for Red Rot Disease Diagnosis

Correct diagnosis of pathogens is the primary requirement in any sound disease management practice. Disease diagnosis and pathogen identification by conventional methods involves isolating the pathogen and characterizing it by inoculation tests (Nithya et al. 2012). As vegetative propagation in sugarcane favors harboring of the pathogen in the planting setts, adequate care should be taken while selecting seed canes. Since, it is difficult to detect incipient infections of *C. falcatum* in seed-pieces, it is recommended to take the planting material from a disease free crop. Three-tier seed production system is being recommended to raise disease free seed cane in the sugar factory area. PCR results confirmed that the primers are able to detect the *C. falcatum* specifically in mixed state of infection in sugarcane which helps to detect the fungal infections more accurately (Viswanathan et al. 2010). The characterization of pathogen has primarily been based upon variation in colour, conidial size and shape, appressoria, colony characters, host association and molecular methods based on polymerase chain reaction (PCR) are frequently being used now a day for detecting fungal pathogens in plant tissues (Bonants et al. 1997; Madan et al. 2000; Wang et al. 2006), due to their increased specificity and

sensitivity in comparison to the traditional techniques. Variability in *C. falcatum* isolates have been characterized by using arbitrary marker system such as RAPD (Madan et al. 2000; Mohan Raj et al. 2002; Suman et al. 2005; Saksena et al. 2013). Kumar et al. (2010) used inter simple sequence repeat (ISSR), universal rice primers (URPs) and RAPD markers to characterize variability among *C. falcatum* isolates. ITS primers were also used to study the variability among pathogen (Malathi et al. 2010). Nithya et al. (2012) developed SCAR markers specific to races and isolates of *C. falcatum* sporulation. But this phenotypic identification is time consuming, expertise specific and not always fully discriminative (Kumar et al. 2010; Abbas et al. 2010).

#### 14.2.1.4 Management of Red Rot

Management of red rot has been a challenging area of work for the pathologists and sugarcane breeders. The epiphytotic of the disease depends upon weather conditions, genotypes, presence of virulent pathogen and time for disease development. These factors must be studied in depth so as to achieve effective control of the disease. It has been observed that once the disease has appeared in the field it is impossible to control. Most of the recommended management practices hence are aimed at prophylactic measures to reduce pathogen build up in the field (Viswanathan et al. 2011). Kumar et al (2012) developed red rot resistance through soma clones, but developed soma clones are loosed originality. Role of endochitinase (ech42) gene in *Trichoderma* was worked out by Singh et al (2013) to prove the efficacy against most dreaded red rot disease of sugarcane caused by *C. falcatum* Went. *Trichoderma* spp. along with *Trichoderma harzianum* strain and *Aspergillus awamori* were tested against the virulent pathotypes of *C. falcatum* (Cf08, Cf09 and Cf401) under *in vitro* and *in vivo* conditions. *T. harzianum* was significantly able to control the incidence of red rot disease in highly susceptible sugarcane cultivar. A single method would not be useful to mitigate the losses from red rot (Agnihotri 1996), hence Integrated Disease Management (IDM) should be practiced. The combinations of different control methods could be adopted for controlling the red rot disease.

#### 14.2.1.5 Resistant Varieties Against Red Rot

Effective control of red rot has been mainly through the use of resistant varieties. Even though genetics of inheritance of red rot resistance is not well established, considerable progress has been made in the production of red rot resistant varieties (Viswanathan et al. 2011). In India, the breeding work is primarily focused on developing red rot resistant varieties (Agnihotri 1996). But as the pathogen is highly variable in nature, therefore, even if a disease resistant variety is released for cultivation, it gets susceptible to red rot disease within 8–10 years because of the development of new more virulent races of the pathogen (Yadav 2006).



#### 14.2.1.6 Chemotherapy and Thermotherapy for Management of Red Rot

A number of fungitoxicants have been tried against the red rot but a little success is found in controlling this devastating disease. This may be because of impervious nature of rind, presence of fibrous nodes at the cut ends, low solubility of fungicides, and lack of broad spectrum fungicides and presence of abundant nutrients in the sett (Agnihotri 1996). Some reduction in red rot incidence has been reported when infected setts were treated with carbendazim and benomyl, which is not commercially produced now for 1–2 h (Anzalone 1970; Chand et al. 1974; Waraich 1983). Soaking of sugarcane setts in 0.25 % suspension of thiophanate methyl and its metabolite carbendazim for 24 h before planting was found to be effective in controlling debris-borne infection (Malathi et al. 2004). In an experiment conducted by Bharadwaj and Sahu (Bharadwaj and Sahu 2014), bavistin showed complete inhibition of mycelial growth of the *C. falcatum*.

Heat therapy has been used for controlling sett borne infection of red rot by various workers. Joshi (Joshi 1954) and Singh (1973) reported complete elimination of sett-borne infection by hot air treatment (54 °C for 8 h). Some workers have used heat and chemotherapy in combination by incorporating the chemical into the hot water tank for adequate control of red rot (Agnihotri 1983). Findings of various workers have proved that moist hot air therapy (MHAT) of seed cane at 54 °C for 4 h (R.H. 95–100) was most effective against red rot (Singh et al. 1980; Dhillon et al. 1983). Anonymous (1980) reported that moist hot air treatment (54 °C for 2 h) was more effective than hot water treatment (50 °C–2 h) in reducing red rot. Aerated steam treatment at 52 °C or the soaking of setts in cold running water for 48 h followed by hot-water treatment (50 °C for 150–180 min) also helps in eliminating the pathogen from infected setts (Singh and Singh 1989).

#### 14.2.1.7 Biological Control of Red Rot

Red rot disease of sugarcane was observed to be biologically controlled through *Trichoderma harzianum* and *T. viride* (Singh et al. 2008). *T. harzianum* and *Pseudomonas spp.* possess the ability to protect the crop from soil borne inoculum of red rot and the efficacy is because of the chitinase enzyme produced by them (Malathi and Viswanathan 2013). Singh et al (2013) clearly supported that *ech42* gene of *Trichoderma spp.* is responsible for controlling the red rot incidence in sugarcane. In an experiment conducted by Singh et al. (2010), addition of SA (salicylic acid) has boosted the protection level significantly against red rot disease and helped *T. harzianum* for inducing systemic resistance in sugarcane. *Ocimum*, *Ginger*, *Onion* and *Garlic* were found to inhibit the mycelial growth. It was also examined that the essential oils, viz. Peppermint oil, Mentha oil, Geranium oil, Patchouli oil and Palmaroza oil were effective in inhibiting the growth of mycelia of *C. falcatum* (Bharadwaj and Sahu 2014). In sugarcane, Viswanathan and Samiyappan (1999a, b) established PGPR (Plant growth promoting rhizobacteria) mediated ISR (Induced systemic resistance) against *C. falcatum* causing red rot disease. The study

conducted by Viswanathan et al. (2005) indicates a possible role of PR (pathogenesis related) proteins in conferring red rot resistance in sugarcane. The peptides MUC1 60 mer and Purothionins mixture significantly inhibited the mycelial growth and spore germination of *C. falcatum* (Edward et al. 2013). It was showed in an experiment that leaf extracts of *Curcuma domestica* and *Datura metel* inhibited the conidial as well as mycelial growth. Smoke of dhup (incense) and tobacco also showed inhibition against conidial germination (Imtiaj et al. 2007).

### 14.2.2 *Smut Disease of Sugarcane*

One of the easily recognized diseases of sugarcane with worldwide distribution. Smut disease is generally observed in Maharashtra, Andhra Pradesh, Tamil Nadu, Kerala and Gujrat. In U.P. its occurrence is noticed in the months of April to June, November and February. A number of varieties were knocked down of cultivation only due to this disease in the past. Smut disease of sugarcane, caused by the fungus *Ustilago scitaminea* (Sydow 1924) leads to considerable yield losses and reduction in cane quality (Ferreira and Comstock 1989) which ultimately affects the sugar industries throughout the world. Smut was first reported from Natal in South Africa in 1877 (McMartin 1945), and thereafter out broke all over the African continent and nearby islands, India, the other south-east Asian countries, the Philippines, Taiwan, and South American countries (Antoine 1961).

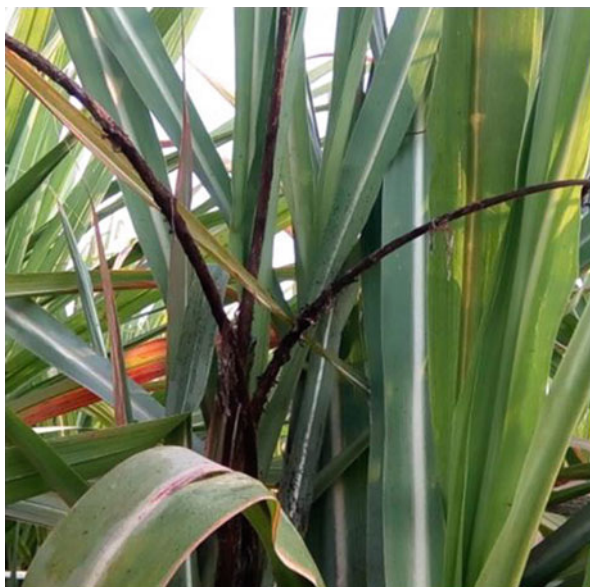
Smut was recorded in 1971 on Hawaii's island of Oahu (Ferreira and Comstock 1989) and in the Caribbean state of Guyana in 1974 (Bates 1975). It then spread to all the Caribbean islands, Central America and mainland USA. In Australia, the disease was first observed in 1998 in a marginal production area on the west coast (Riley et al. 1999) and was detected in the main producing area (east coast) in 2006. Only Papua New Guinea which is considered as the center of origin of *Saccharum officinarum*, the main species involved in modern cultivars, and Fiji Islands are still free of this disease.

#### 14.2.2.1 *Symptomology and Epidemiology*

This is also a sett borne disease. The leaves of the crown become shorter and pointed making a fan like structure of the top. The affected plants become thin with elongated internode. A whip like structure emerges out from the top which ranges from 6 in. to 1 m. The black colored whip is initially covered with a silvery membrane which ruptures later to disseminate the spores in air for secondary infection on other plants (Viswanathan et al. 2000) (Fig. 14.2). The first appearance of the apical whips is seen around 120 days of planting. The second flush of whip emergence produces an enormous quantity of teliospores which infect the terminal and lateral buds in the rapidly growing crop. The infected buds may remain dormant and may germinate to produce lateral whips in the third flush of whip production. The



**Fig. 14.2** Smut symptoms in field crop of sugarcane



infection producing the third level of whips is believed to be critical in the epidemiology of the disease (Ramesh Sundar et al. 2012). Smut is generally favored by hot dry weather conditions. Plant stress increases frequency of whip development. Under high stress conditions the cultivars may show symptoms, otherwise they normally do not produce whips. Teliospore survival is decreased rapidly by soil moisture. Similarly, high rainfall reduces the severity of smut development. In subtropical areas, after severe winter the level of smut decreases probably due to death of smut infected plants. Disease severity usually increases in the ratoon crops (Viswanathan and Rao 2011).

#### 14.2.2.2 Molecular Markers for Smut Disease Diagnosis

Smut is morphologically diagnosed by the presence of different types of whip such as long, closed, twisted, short and multiple whips. Smut infected sugarcane clump also shows profuse tillering and poor cane formation. Some unusual symptoms of smut infection are apical deformity, floral infection, malformed spindle, and bud proliferation (Ramesh Sundar et al. 2012).

The polymerase chain reaction (PCR) techniques, including PCR and nested-PCR are widely used as rapid, specific and sensitive tools for the detection and evaluation of plant diseases (Kamel et al. 2003; Farid et al. 2006; Kawther 2008; Li et al. 2009; Liu et al. 2009). For detection of *U. scitaminea*, so far only one PCR-based method was reported, which was based on the b loci (bE4/bE8E) mating-type gene sequence of *Ustilago maydis* (Albert and Schenck 1996). Nested-PCR detection offers higher sensitivity in detection of plant diseases as compared to PCR-based

methods (Li et al. 2009; Liu et al. 2009). The life cycle of the fungus is regulated by *a* and *b* mating-type loci within the sporidia. Mating type *a* has two alleles which encode a pheromone and a receptor whilst *b* is multi allelic and appears to control pathogenicity and sexual development (Bakkeren et al. 1992). By using primers based on the *U. maydis* *bE* mating-type gene, Albert and Schenck (Albert and Schenck 1996) sequenced the corresponding gene in *U. scitaminea*. Since then molecular detection of the smut pathogen has become possible using PCR to amplify the *bE* mating-type gene of *U. scitaminea*. Based on nucleotide differences in the internal transcribed spacer (ITS) sequences of *U. scitaminea*, a pair of species-specific primers, SL1 (5'-CAGTGCACGAAAGTACCTGTGG-3') and SR2 (5'-CTAGGGCGGTGTTTCAGAAGCAC-3') was designed by using a panel of fungal and bacterial species as controls. The primers SL1/SR2 specifically amplified a unique PCR product about 530 bp in length from *U. scitaminea* strains with a detecting sensitivity at 200 fg of the fungal genomic DNA in a 25 µl reaction solution. To increase sensitivity, a nested-PCR protocol was further established, which used ITS4/ITS5 as the first-round primers followed by the primer pair SL1/SR2. This protocol increased the detection sensitivity by 10,000-fold compared to the PCR method and could detect the fungal DNA as low as 20 ag (Shen et al. 2012).

Amplified Fragment Length Polymorphism (AFLP) fingerprinting technique was used to estimate the degree of genetic variation among 38 isolates of sugarcane smut collected from 16 sugar industries in 13 countries by Braithwaite et al. (2004). The AFLPs revealed a low level of genetic variation among 31 isolates, however, seven isolates collected from Taiwan, the Philippines and Thailand, were genetically more divergent. There was no significant difference between isolates collected in 1984 and those collected 16 years later from the same sugar industry, suggesting that genetic change occurs slowly in this fungus and very limited changes have occurred in the existing populations. The induced resistance mechanisms in sugarcane in response to challenge by the pathogen *U. Scitaminea* was investigated using the cDNA-AFLP differential display technique by Thokoane and Rutherford (2001).

Smut disease on sugarcane though first recorded in South Africa in 1877 on the *Saccharum sinense* clone known as 'China cane', it was most likely present in Asia for much longer. There are circumstantial reports that smut has always been present in India (Antoine 1961). Indian sugarcane (*Saccharum barberi*) is very susceptible to sugarcane smut and the wild grass *Saccharum spontaneum* is considered to be a collateral host of sugarcane smut and a major reservoir of inoculum in India (Chona 1956). An Asian origin for *U. scitaminea* may explain the higher level of genetic variation among the Asian smut isolates. Further sampling from Asia, particularly India, and also from relatives of sugarcane (e.g. *S. spontaneum*, *S. barberi*, *S. sinense*) may reveal more variation within the Asian region. Interestingly, the centre of origin of *Saccharum officinarum* is believed to be New Guinea (Daniels and Roach 1987), and this region remains free of smut even to this date. Alternative fingerprinting techniques, such as simple sequence repeats (SSRs or microsatellites), may provide higher sensitivity and generate more polymorphisms to allow to verify the significance of the weak clusters being observed, or to reveal the existence of yet other clusters.

### 14.2.2.3 Role of Different Compounds in Resistant Varieties Developments

The resistance to a disease occurs by accumulation of different compounds such as: phytoalexins (i.e. low molecular mass antimicrobial compounds that accumulate at sites of infection); systemic enzymes that degrade pathogens (e.g. chitinases,  $\beta$ -1,3-glucanases and proteases); systemic enzymes that generate antimicrobial compounds and protective biopolymers (e.g. peroxidases and phenoloxidases); biopolymers that restrict the spread of pathogens (e.g. hydroxyproline-rich glycoproteins, lignin, callose); and regulators of the induction and/or activity of defensive compounds (e.g. elicitors of plant and microbial origin, immune signals from primed plants and compounds, which release immune signals) (Kuc 1990). Three varieties of sugarcane have been studied for the production of glycoproteins after smut infection viz., Barbados which is extremely susceptible; Jaronu shows moderate resistance, while Mayari is extremely resistant to smut. When vegetative buds of sugarcane was inoculated with teliospores and cultured under field conditions for 12 months, it was observed that soluble polysaccharides and glycoproteins of juices increased after infection in Jaronu and Mayari plants, but the same decreased in Barbados specimens (Martinez et al. 2000). Several glycoproteins, defined as mid-molecular mass (MMM) or high molecular mass (HMM) macromolecules are produced as a primary response to smut infection (Legaz et al. 1998a, b). Teliospore germination in the presence of both MMM and HMM decreased about 50 % following 5 h of teliospore contact with glycoproteins which may be related to the ability of glycoproteins to produce cyto-agglutination. Binding of fluorescein-labelled glycoproteins was studied by fluorescence microscopy, which reveals that staining of cells was not even, but mainly restricted to the contact zone between two individual teliospores when aggregated (Fontaniella et al. 2002). Ungerminated fungal spores generally lack specific localization of their organelles and most are apparently able to germinate from any point. Hence, one of the major changes that must occur during germination is the establishment of cytoplasmic polarity (Fontaniella et al. 2002). Thus disease resistance seems to be a multifactorial process.

### 14.2.2.4 Management of Smut Disease

#### Breeding and Selection

Breeding for resistance to sugarcane smut, caused by *U. scitaminea* H. & P. Sydow, is a feature of sugarcane breeding programs in all countries where the disease is present. The disease can be successfully managed with resistant varieties (Comstock 2000). Breeding a new variety starts with the crossing of parent clones to produce sexual seed. In 2004, the smut breeding strategy was reviewed and a new strategy was developed that targeted 25 % of crosses with mid-parent ratings in the resistant range and at least 25 % in the intermediate range. This new strategy was almost achieved in 2005

and was exceeded in 2006 in Australia. Overseas research has shown that, even in crosses between two highly susceptible parents, between 6 and 20% of progeny will be smut-resistant (Wu et al. 1983; Chao et al. 1990). These resistant progeny from susceptible crosses will be assessed as potential varieties and will be used as parents. The methods for screening for smut resistance have been reviewed by Ferreira et al. (1980). Screening is conducted in most countries by dipping setts in a suspension of smut spores and rating clones on the percentage of infected plants in a plant and ratoon crop. Most researchers recommend that the spore suspension should contain approximately  $1 \times 10^6$  viable spores per mL. Collecting sufficient spores to conduct large screening trials can be difficult when heavily infested crops are not common. When spores are in short supply, painting a spore paste or spraying spores onto buds are alternative methods. Maintaining the inoculated setts in a warm (31 °C), humid environment provides the ideal conditions for infection (Bock 1964). There are no commonly accepted methods for screening true seedlings for smut resistance. A number of methods have been published but none have been widely adopted.

### Integrated Pest Management

IPM is a tactic strategy for minimizing the damage caused by the diseases which vary according to the variety of modifying factors (Kendrick 1988). IPM emphasizes on the combination of methods in contrast to single strategy control for minimizing side effects (Allsopp and Manners 1997). Pathogens are regarded as indicators of unmanaged design system for control of disease rather than an enemy in an IPM system.

### Application of Fungicides

The use of resistant varieties has proved to be an efficient and economic method for the control of sugarcane smut (Villalon 1982; Pruett and Waller 1989; Phelps and Donelan 1991). However, the development of resistant cane varieties takes time to achieve, so fungicides are applied for short term. Infection due to *U. scitaminea* is through air-borne spores which gain entry into buds of standing canes and from spores in the soil or irrigation water that enter buds of planted setts (Bebbee 1988). As the bud sprouts, the fungus grows within the resulting seedling and arrests its growth by turning the terminal axes into its own reproductive structure—the sorus—which appears when the cane is 2.5–5 months old (Antoine 1961; Peros 1984). Systemic fungicides such as benomyl and carbendazim (Gul and Hassan 1989; Olufolaju 1993) as well as several protectant fungicides such as mancozeb, chlorothalonil and captan have also been applied by soaking the setts in their solutions or suspensions (Kidd and James 1991). The plant sets are dip-inoculated in smut spore suspension and then in solutions or a suspension of six fungicides for 10 min and planted out in glasshouse and field trials. Mancozeb, carbendazim + maneb, metalaxyl + carboxin + furathiocarb, pyroquilon, benomyl and

chlorothalonil are all effective against *U. scitaminea*. However, the best disease control was obtained with pyroquilon at 4.0, carbendazim+maneb at 4.57 and metalaxyl + carboxin + furathiocarb at 9.9 g a.i.l<sup>-1</sup>, respectively (Wada 2003).

### Hot Water Treatment (HWT)

The hot water treatment of 50–55 °C for 2 h has been reported for the control of ratoon stunting disease (RSD), and other combinations of time and temperature, also control smut disease in sugarcane (Srinivasan 1971; Steiner and Byther 1971) in Rhodesia. Thompson HA (1973, personal communication) in Mocambique observed the development of smut in canes that were previously hot water treated (HWT), and revealed higher incidences of disease in first and subsequent ratoon crops of smut susceptible varieties compared to non-HWT canes of the same varieties. So, hot water treatment of seed cane for the establishment of RSD-free nurseries was discontinued in Rhodesia, where smut was endemic. The major commercial variety in Rhodesia is NCo376, which- is very susceptible to smut. James (1974) also confirmed a greater incidence of smut in the ratoon crops-of both heat-treated compared to RSD-inoculated cane and healthy compared to RSD-infected cane in Rhodesia and suggested an interaction between the RSD bacterium and *Ustilago scitaminea*, in cane infected by both pathogens, whereby smut is suppressed. In Zambia also demonstrated a greater incidence of smut in the variety NCo310 after HWT.

## 14.2.3 Wilt Disease of Sugarcane

Among the disease of the sugarcane, wilt disease has also received the attention recently by the researchers working on sugarcane. Serological characterization has been made by the several workers across the world but the information at molecular level is scanty. Limited reported has been made by the workers especially in India (Viswanathan et al. 2010). This pathogen has been reported throughout the world in various research publications. The disease is still spreading in various sugarcane producing areas of the world and now a days it is being judged and counted as the second major disease of sugarcane in India.

### 14.2.3.1 Epidemiology and Symptoms of Disease

Wilt of sugarcane is a serious stalk disease in India and other South Asian countries. Although wilt has been associated with red rot in causing severe damage in different countries, wilt alone causes severe damage to certain sugarcane varieties in India (Viswanathan and Padmanaban 2008). However, conflicting claims have been made regarding the true causal organism of sugarcane wilt as species of *Fusarium*,

*Cephalosporium* and *Acremonium* by different authors (Agnihotri and Rao 2002) and information on the pathogen(s) involved and its variation, are not clearly brought out. The pathogens associated with the disease were recorded to be different from area to area and between investigators. Butler and Khan (1913) were the first to describe the disease in India and they identified the pathogen as *C. sacchari*. A similar disease named as “*Fusarium sett*” or “Stem rot” was recorded by Bourne (1922) in Barbados. Abbott (1932) also observed the disease in Louisiana. Gams (1971) who reviewed the taxonomy of *C. sacchari* also considered it as *F. moniliforme* var. *subglutinans* and coined a new combination as *F. Sacchari* (Butler) W. Gams to which both *C. sacchari* and *F. moniliforme* var *subglutinans* were made synonyms. Later, Nirenberg (1976) distinguished two varieties of *F. Sacchari* i.e., *F. sacchari* var *sacchari* and *F. sacchari* var *subglutinans*, the former having mostly unseptate conidia in the aerial mycelium, no sporodochia, while the latter with one to three septate conidia, macroconidia more commonly formed in sporodochia. Wilt incidence is always higher in ratoon crops compared with the plant crop. Besides yield reduction, wilt also causes 15–30% reduction in juice extraction and up to 20% in sugar recovery. The pathogen is primarily transmitted through infected seed canes in the field. Also the fungus surviving in sugarcane debris in the soil serves as a source of inoculum to infect sugarcane from soil. Symptoms of wilt appear in monsoon with the yellowing of the crown. Later on, the weight of cane starts to become lighter due to increase in pith area. From October onwards, the whole crown withers away and the cane becomes hollow. After splitting longitudinally, light pink to dark purple coloured tissues are seen at nodal area along with the presence of grayish mycelium in the hollow internodes. The dried internodes remain intact at the nodes and do not detach easily from the nodes (Agnihotri and Rao 2002) (Fig. 14.3). The wilt fungus can survive in soil for 2.5–3 years. Secondary spread from field to field occurs through rain and irrigation water. Wilt is very common in certain locations where conducive environment and susceptible hosts are available. However, the disease expression and its severity are being influenced by various biotic and abiotic factors in the field. Although abiotic factors play a decisive role on disease severity, biotic agents facilitates entry of the pathogen inside the root or stalk. Hence, role of biotic factors are of paramount importance in wilt initiation, development and severity. Wilt is more dangerous and causes enormous damage to crop in association with red rot, pineapple disease, borers or scale insects. Wherever root borer, *Emmalocera depressella*, infestation is severe, more wilt infection is noticed. However such borer-wilt complex was rarely noticed in the tropical region except, Gujarat state. In this case nearly 80–90% borer infestation resulted in about 60% wilt infection (Viswanathan and Rao 2011).

The disease symptoms appear during the monsoon and post monsoon periods, affected plants, either singly or in small groups, displays the disease symptoms. This is followed by yellowing and/or withering of upper leaves. The midrib of all leaves in a crown generally turns yellow, while the leaf lamina may remain green. Towards the end of the crop growth, the cane becomes light and hollow and dry. On splitting up on the affected canes, at the early stage of the infection, the diffused purple or muddy red colour is seen as conical patches on each node just above the



**Fig. 14.3** Sugarcane stalk (LS) showing the symptoms of Wilt disease



growth rind. Occasionally, above the badly affected internodes are seen one or two red stripes, (vascular strands) that pass from one internodes to another. In severe cases, the spindle shaped cavities, tapering towards the nodes; develop in each internode because of general recession and rapid desiccation of tissues. Sometimes cavities also develops in the nodal tissues, this makes the canes tubular. In severe cases, affected stools withered and dry away.

#### 14.2.3.2 Managements of Wilt Disease

Whenever severe wilt epidemics occurs and devastated the cane cultivation during the past, replacements of varieties with disease resistance were done and sugar industry was saved from this threats in India. Hence, growing resistant varieties are the cost effective and it will assure sustained cane cultivation in a region. So far, there is no systematic screening programme for screening wilt resistance, as it is difficult to reproduce the symptoms by artificial inoculation unlike red rot and smut which have different methods for screening. Viswanathan et al. (2009) have identified effective biocontrol isolates of *T. harzianum*, *T. viride* and *T. pseudokoningii* from sugarcane rhizosphere. Detailed field trials to manage the disease revealed that soil application of *Trichoderma* multiplied on press mud is effective in managing wilt in endemic locations of Gujarat and Andhra Pradesh. However, more trials needs to be conducted to validate the results in different endemic locations in the country (Viswanathan and Rao 2011).

### 14.2.4 Pineapple Disease

This is a soil borne disease. Due to smell like pineapple from the diseased setts, the name of disease was given as pineapple disease. The odor comes out from the diseased setts due to the formation of ethyl acetate because of metabolic changes. In India, this disease has been reported from Punjab, Maharashtra, Karnataka, Tamil Nadu, Kerala and U.P. Sett rot or pineapple disease, caused by ascomycetous fungus, *Ceratocystis paradoxa* (de Seynes) Moreau, is one of the sett borne and soil borne disease of the sugarcane. The symptoms associated with this organism include a smell of matured pineapple fruit, attributed to the production of ethyl acetate by the metabolic activity of the fungus (Coale 1989). The disease has been reported from many of the sugar cane growing regions with heavy losses under severe condition. The crop is propagated by means of stem cuttings. The disease mainly affects the germination of the setts at early stages of planting and the most serious losses are through the failure of infected cuttings to germinate (Comstock et al. 1984), although standing cane may also become infected (Manzo 1975; Natarajan and Subba Raja 1976). In India the disease was reported from different varieties of the crop (Singh et al. 1990).

#### 14.2.4.1 Symptoms and Development of the Fungus

The symptoms of disease appear 2–3 week after planting. The pathogen gets entry through the cut ends of the setts. The affected setts become red and there after the colour changes to blackish brown. The internodes of the sett become hollow. The pathogen grows and multiplies rapidly. The setts ultimately are rotted away and the buds do not germinate (Fig. 14.4). The disease is also noticed with association of white leaf disease (phytoplasma) in India at few places (Tiwari et al 2015b). Major factors that affect the growth and proliferation of the fungus *Ceratocystis paradoxa*, are soil moisture level, pH, relative humidity (RH), carbon and nitrogen sources as well as existing interactive microflora in the soil (Adiver 1996; Yadahalli et al. 2006, 2007). The soil moisture level of 60 % favoured significantly highest sett germination (65.60 %) followed by 80 and 40 % soil levels (55.30 and 46.70 % germination, respectively). The 100 and 20 % soil moisture levels recorded significantly low set germination (28.60 and 38.50 % germination, respectively). Soil moisture level of 60 % led to least development of sett rot (5.5 %). Increase or decrease in soil moisture above 60 % led to the increase in sett rot development. The interactions between days after planting and soil moisture status were also significant (Yadahalli et al. 2006). Soil moisture is growth determinant for most of the soil dwelling organisms. It is a medium through which the soil fungi absorb their nutrients and carryout most of their metabolic and physiological functions (Bolkan et al. 1982). Disease incidence was more in sterile soil than the unsterile soils, as the pathogen has to compete with other soil microbes for nutrition and space. The results are in conformity with the findings of Hooda and Grover (1988). The carbon is essential ingredient of all the living organisms and almost half of the dry weight of the fungal cell

**Fig. 14.4** Sugarcane stalk (LS) showing the pineapple disease symptoms in sugarcane



consists of carbon. Carbon compounds are equally important in fungus nutrition (Cochrance 1958). Starch induces highest growth of *C. paradoxa* with maximum biomass of 1039.99 mg as well as sporulation (79.67 spores/mL) followed by glucose (72.30 spores/mL). The lowest sporulation ( $7.67 \times 10^6$  spores/mL) is recorded in treatments with inositol as carbon source. However, treatments with the maltose and xylose as carbon sources were on par and non-significant (Yadahalli et al. 2006). Among nitrogen sources, nitrate forms have been reported to be excellent sources for imperfect fungi (Suryanarayanan 1958). *C. paradoxa* produce significantly highest biomass (1590.70 mg) in potassium nitrate, ammonium nitrate (1496.97 mg), ammonium oxalate (1405.00 mg) and magnesium nitrate (1234.30 mg) were next best preferred sources. Colony growth and conidial formation are sharply affected in different levels of pH. The growth of *C. paradoxa* is maximum in pH 6.5 (82.67 mm) irrespective of media. Similarly, the highest mean growth is recorded in PDA (62.75 mm) irrespective of pH levels tested. Lowest sett rot incidence is recorded at 60% RH at all the intervals (Yadahalli et al. 2007). These results concur with the findings of Laxminarayana (1981).

#### 14.2.4.2 Mechanism of Inhibition of Bud Germination

Pineapple disease infected sugarcane tissue may contain ethyl acetate up to 1% that is sufficient to inhibit germination of buds (Kuo et al. 1969). The disease can reduce the sett germination up to 47% (Anonymous 1999) which subsequently hamper the

yield of cane and the yield reduction may be as high as 31–35 % (Anonymous 2000). Invading pathogens often produce certain compounds inside host that alter the metabolism, paving a way for further proliferation and growth of the pathogen. Rotting of sugarcane setts after planting often is a result of certain toxins produced by the invading pathogen, causing poor stand of the crop. The major cause of rotting of sugarcane setts is invasion by *Ceratocystis paradoxa*, an ascomycetous fungus, causing sett rot disease. The volatile toxic substance was identified as ethyl acetate by the experiment conducted by Yadahalli et al. (2007). In the inoculated setts, there was no germination and radical initiation and in treatment involving ethyl acetate, the bud germination was completely inhibited but only few roots (15 roots/node were observed). The studies conducted by Flori Bai (1985) also revealed similar results. To confirm the release of ethyl acetate (the phytotoxic volatile substance) by the pathogen, a hydroxylamine acid test was conducted in which 1 ml of 0.5 M hydroxylamine hydrochloride in 95 % ethanol and 0.1 ml of 6 M aqueous NaOH was heated to boiling. Slightly cooled and to this 2 ml of 1 M hydrochloric acid was added. The appearance of magenta colour by the addition of one drop of five per cent ferric chloride indicated the presence of ethyl acetate (Flori Bai 1985).

#### 14.2.4.3 Management of Pineapple Disease

Pineapple disease can cause significant reductions and the most serious losses are through the failure of infected seed pieces to germinate (Wismer and Bailey 1989), although standing cane may also become infected (Manzo 1975; Natarajan and Subba Raja 1976). In Florida, two cultivars of commercial significance, CP 74–2005 and CP 72–2086, are highly susceptible to this disease (Coale 1989), these two cultivars together accounted for 10.3 % of the Florida sugarcane hectareage in 1989 (Coale 1989). On the organic soils of the Everglades Agricultural Area (EAA), disease development appears to be most favoured by cool, wet soil conditions (Coale 1989). Although Florida's dry season generally coincides with the cool winter months (December–February), heavy localized rainfalls during this period are not uncommon. Stand reductions due to pineapple disease of severities sufficient to require selective replanting have been observed in poorly drained fields in the EAA. A number of control measures have been recommended for pineapple disease, including the use of resistant cultivars, site selection, increased seed piece length, and avoidance of factors which down the seed piece germination (Wismer and Bailey 1989). In many cane-growing regions of the world, seed pieces are routinely treated with a fungicide as part of the planting operation (Steindl 1970; Comstock et al. 1984; Taylor and Ryan 1984; Sivanesan and Waller 1986). In Florida, where seed piece treatment is not practiced, seed pieces are typically overlapped and at least doubled in the furrow at planting. This practice is to insure establishment of an adequate number of primary shoots. When a pineapple disease susceptible cultivar is being planted under conditions favorable for disease development, the seeding density may be further augmented. Although pineapple disease is widespread in Hawaii (Rashid and Trujillo 1974) and Australia (Aberdeen 1969),

the standard planting procedure involves only a single line of fungicide-treated seed pieces per row. This seeding rate, in combination with seed piece treatment, is capable of producing stands sufficient for maximizing yields.

Biocontrol of plant pathogen involves the use of biological processes to reduce the inoculum density of pathogen and to maintain their soil population below the disease threshold level. The pathogen suppression in the soil is considered as an important step in the control of disease as it involves the direct disinfestations of the soil. The pathogen in the absence of their hosts survives either as dormant propagules or actively as saprophytes on dead organic substrates of the host in the soil (Mahalingam et al. 2011). The antagonistic activity of some soil fungi against *C. paradoxa* has been investigated in vitro dual culture and with cell free culture filtrates of fungi amended in medium. The toxic metabolite produced by the initial fungal colonies of natural substrate may act to slow or prevent invasion by other species (Ambikapathy et al. 1994). *Trichoderma* sp. is most common fungal biological control agents that have been comprehensively researched and deployed throughout the world.

### 14.2.5 Pokkah Boeng Disease

Pokkah boeng is now playing a very important role and spreading rapidly in UP. Vishwakarma et al. (2013) showed increasing trend of disease incidence and most of the commercial cultivars affected by the disease ranged from 1 to 90%. Although pokkah boeng comes under minor concern but these days it is going to be major on basis of their rapid epidemiology during last few years. Nowadays, the incidence and severity of pokkah boeng disease has been reported from major sugarcane growing states like Uttar Pradesh, Maharashtra, Punjab, Haryana, Assam, Tamil Nadu and Bihar in India and other sugarcane growing countries. This disease appears with the onset of the monsoon. Broad leaf cultivars are more vulnerable to get infection by this disease. The incidence of this disease reduces with the offset of the monsoon itself. The crown leaves of the affected plant become pale yellow or white which later on turn to dark reddish brown due to rotting and are detached from the top. Thus the affected plant looks without top and becomes thin (Sharma and Kumar 2015).

#### 14.2.5.1 Epidemiology

Pokkah boeng of sugarcane caused by *Fusarium moniliforme* and the pathogen was first described by Sheldon (1904) and the perfect stage of pathogen is *Gibberella fujikuroi* (Sawada). *Fusarium* now confirm by several worker as a causal agent of pokkah boeng in sugarcane in Asia as well established pathogen by many worker (Singh et al. 2006; Govender et al. 2010; Mohammadi et al. 2012). Pokkah boeng is the serious and devastating disease not only in central Uttar Pradesh but also in the whole of the Southern and Northern sugarcane growing zone of India. Pokkah

boeng is a Javanese term denoting a malformation or distorted top was originally in Ja1 va but in that time no causal agent was established and its incidence was recorded by Padwick (1940). Pokkah boeng disease recorded in all over the countries where sugarcane grown and pathogen spreads in wind-blown rain, infected cane cuttings, pupae and adults of sugarcane stem borers (Patil and Hapase 1987; Whittle and Irawan 2000). This disease was well-known in sugarcane for long time but severity of disease reported in two commercial varieties Co7219 and Co C671 in Maharashtra 1983–1984 (Patil and Hapase 1987). *Fusarium moniliforme* var. *subglutinans* reduce the quality of the harvested crop mainly among varieties with high sugar yields the sugar production depending upon the variety up to 40.8–64.5% (Duttamajumder 2004).

The *pokkah boeng* pathogen also attacks sorghum and had been reported that the disease was caused by *F. moniliforme* (*Gibberella fujikuroi*) (Mohammadi et al. 2012), now known as *F. verticillides*. Fungus infects a wide range of species including monocotyledons and dicotyledons causing various diseases such as seedling blight, scorch, stalk and root rot, abnormal stunting or hypertrophy. Pathogen of *Pokkah boeng* disease has been associated with several diseases of sugarcane such as sett rot, root rot (Verma et al. 1984), wilt (Saponaro 1980; Waraitch and Kumar 1982) and knife cut (Zummo 1970; Kamal and Singh 1979).

#### 14.2.5.2 Symptomology

The characteristics symptoms of *pokkah boeng* disease are the appearance of chlorotic patches towards the base of the young leaves, in acute cases disease shows distortion of stalk with external and internal cut like lesions and rotting of apical part of stalk. Under field conditions, the disease may develop many variations from the general symptoms, but the final result is usually a malformed or damaged top and stalk. The base of affected leaves is often narrower as compare to normal leaves. Knife cut symptom of the disease reported in varieties CoS767, CoC671, CoC8014, Co1158, CoS8315 and CoS 8436 (Patil 1995). Development of disease symptom in four phases namely chlorotic phases I & II, top rot and knife cut phase. The apical leaves may also show pronounced wrinkling and twisting depending upon the susceptibility of varieties and existing climatic conditions also malformed or damaged top and stalk due to this disease (Martin et al. 1961). The symptoms of *pokkah boeng* disease were mainly two types i.e. chlorotic phase and acute phase of top rot (Patil and Hapase 1987) and Knife cut (fourth phase) of this disease in Maharashtra (Patil 1995). Similar findings of about the disease symptom were also reported by workers (Eira et al. 1974; Higgy et al. 1977).

Major and minor (N, P, K, S, Zn, Fe, Cu and Mg) study of the sugarcane plant affected by the *pokkah boeng* showed the decreasing pattern in stalks and leaves of diseased plant as compare to healthy ones and its incidence have been noticed on broad leaves varieties of sugarcane during monsoon. This may be leads to the reduction in weight of cane, length of internodes, juice percent, girth, pol per cent and total sugars in juice in infected canes of varieties CoS 8436 and CoS 88230 (Singh et al. 2006).



### 14.2.5.3 Pathogenesis

The pathogen enters into the host tissues through any injury by insects/borers or natural growth cracks etc. After the entry of pathogen, the infection thread develops normal hypha which grows within the host tissues for some time and then emerges out through the cells to the outer surface and develops acervuli. Rains and heavy dews usually wash the acervuli developed on nodes and internodes and the spores get lodged around the nodes behind the leaf sheath. The spores germinate and the mycelium gets established in bud scales, root primordial or leaf scars and later within the plant tissues. Electron microscopic of infected leaves revealed that after lodging of conidia and incubation of minimum 1 month at the time of germination, the thin walled bulliform cells of the epidermis are attacked and soon collapse then the older cells of the epidermis are attacked (Patil 1995). From the epidermal cells, the hyphae enter the underlying tissues. Changes in the structure of stomata were also observed in the infected leaves but there was no evidence found for the entry of pathogen through stomata.

### 14.2.5.4 Molecular Detection of Pokkah Boeng Pathogen

PCR-based methods offer rapid and cost-effective tools of molecular diagnosis. Real-time PCR coupled with conventional PCR techniques can reliably detect and quantify pathogens in the fungal isolates or from infected leaf samples (Sayler et al. 2012). Several PCR assays have been developed to identify species of *Fusarium* based on the translation elongation factor gene (*tef1*) (Sampietro et al. 2010; Wang et al. 2011; Amatulli et al. 2012), polyketide synthase gene (*pks*) (González-Jaén et al. 2004), and trichodiene synthase gene (*tri5*) (Zhu et al. 2010). The sequence of the ITS-rDNA is highly conserved but sufficiently variable among species of *Fusarium* species complex. The ability of the ITS region to differentiate and provide accurate and rapid detection of fungus has been reported at the species-level (Zeng et al. 2007). Specific PCR primers based on ITS sequences for *Fusarium* species complex have been designed based on two single nucleotide polymorphisms (SNPs) present in the ITS region (Dita et al. 2010). Lin et al. (2014a, b) developed rapid, efficient, and specific PCR assay to detect pathogens of sugarcane pokkah boeng based on the differences in the ITS-rDNA sequences between two species of *F. verticoides* and *F. proliferatum*. Species-specific primers and probes were used for their specificity and sensitivity. These primers were evaluated to detect *F. verticoides* and *F. proliferatum* using purified DNA from 84 isolates or 28 infected sugarcane leaves. TaqMan PCR using FAM-gx1 and FAM-gx2 primer–probe combinations was very sensitive, with a detection limit of 10 pg/μl fungal DNA. Conventional PCR using Pgx1 and Pgx2 primers, which also targeted the ITS-rDNA region, resulted in fragments at the size of 439 bp with Pgx1 and 400 bp with Pgx2. However, the detection limit for the conventional PCR was approximately 10 ng of fungal DNA input.

Lin et al. (2014a, b) used the sequences of the internal transcribed spacer from 103 isolates were amplified by PCR using fungus-conserved ITS1 and ITS4 primers and sequence alignment to studies the diversity and differences among the isolates

of *Fusarium* species complex. Two pathogens (*F. verticoides* and *F. proliferatum*) to cause sugarcane pokkah boeng in China were identified by morphological observation, pathogenicity test, and phylogenetic analysis. Both conventional PCR and TaqMan PCR were developed to reliably detect the species of *Fusarium* using purified fungal DNA as input based on their ITS sequence. The TaqMan PCR was a more efficient tool for the early diagnosis of disease, and it was capable of detecting pokkah boeng in symptomatic young tissues of infected plants. They also found similar symptoms to sugarcane pokkah boeng, such as the twisted and curling symptoms of crown leaves, and was found to be infected by *Phoma* sp. based on the morphological characteristics and the ITS sequence of rDNA (Lin et al. 2014a, b). TaqMan PCR is more sensitive method to diagnose pokkah boeng pathogen which could not be easily detected in most of leaf tissues by conventional PCR.

#### 14.2.5.5 Control Management

Basically it is an air-borne disease and primarily transmitted through the air-circulation and secondary infection takes place through the infected setts, irrigation water, splashed rains and soil. The pathogen (*F. moniliforme*) can survive for 12 months in the plant debris under natural conditions and can remain viable for more than 10 months under laboratory conditions. Spraying of different fungicides like Bavistin (1 g/l of water) or Blitox (0.2 %) or Copper oxychloride or 0.3 % Dithane M-45 (3 g/l of water) are the effective for reducing the pokkah boeng disease (Kamal and Singh 1979). Two to three sprayings with an interval of 15 days interval reduces the multiplication of a pathogen and losses in yield and quality of cane and therefore, paired row or wider spacing planting of sugarcane is necessary to facilitate the plant protection operations. Canes showing 'top rot' or 'knife cut' should be rouged out from the fields as they are shown. Planting of healthy seed material/use of resistant varieties and follow of Integrated Disease Management practices are the best way to prevent disease incidence (Patil et al. 2007). *F. moniliforme* can be disseminated horizontally by airborne spores or crop debris and vertically through the seed pieces. So it is important to uses of resistant varieties and fungicide applications. Both the processes for controlling, is limited and there is increasing need for novel and environmentally sound strategies to control this and other diseases of sugarcane. *Burkholderia* isolates from sugarcane plants is a crucial step toward further development of these isolates for biological control of Pokkah boeng and other sugarcane diseases. The endophytic bacteria community associated with sugarcane harbours multiple genera with potential for plant growth promotion and disease control (Mendes et al. 2007).

#### 14.2.6 Leaf Binding Disease

Leaf binding disease of sugarcane was first observed in 1918 in Argentina (Abbott 1964) caused by *Myriogenospora aciculispore* V & *Fusarium moniliforme* (*F. verticoides*) and it was called *Myriogenospora* leaf binding disease or Tangle top disease.



**Fig. 14.5** Symptoms of *Pokkah Boeng* of sugarcane (a) Leaf (b) Shoot

In India, leaf binding disease was identified by biological, microscopic and molecular characteristics and also attempted the control prospective of this disease by utilizing fungicides. The disease falls in the minor category and is not prevalent in Asian countries. The occurrence of the disease is in monsoon period (June last week to September) which is the grand growth period of the host plants (Tiwari et al 2016).

#### 14.2.6.1 Symptomatology

Mechanical binding with necrosis of leaves is major symptoms of the disease. Early death of infected shoots and the dwarfing and fasciation is also found. The fasciation occurred due to mechanical binding by fungus tissues of the adjacent leaves and clumps during the growth of the shoots. No characteristic symptoms found on the stalk of the sugarcane plants (Fig. 14.5). The growth of the affected plants completely checked due to the disease and thus reduction in yield was observed around 2 % in all the affected varieties (Tiwari et al 2016).

### 14.3 Conclusion

The fungal diseases of sugarcane are responsible for the quality and stability of the crop yield. Being as long durational crop and its vegetative proppogation, monoculturing and rationing practices makes it easily prone to build up of the diseases.

Hence, an integrated approach involving cultural, chemical/physical methods, host resistance and legislative measures is suggested for the sustainable management of sugarcane diseases. Periodical monitoring and scouting of the disease population estimates are very essential and basic pre-requisites for deciding the need based control measures and to develop IDM. This is the key for successful implementation of IDM. Such survey and surveillance should be regular so as to know the changing disease situations due to agro- climatic factors. Molecular breeding approaches help in developing resistance varieties. Early disease diagnosis and marker development helps in detection and developing management strategies for the diseases. Genomics, transcriptomics and proteomics approaches need to further explore to understand the exact mechanisms of host-pathogen interaction and involvements of different up-regulating and down regulating gene involvements.

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